

AS • Edexcel • Physics

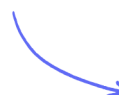
 32 mins  3 questions

Structured Questions

Waves, Electrons & Photons

Diffraction / The Diffraction Grating Equation / Core Practical 8: Investigating Diffraction Gratings / The Wave Nature of Electrons / The de Broglie Equation / Transmission & Reflection of Waves / Pulse-Echo Technique / Wave-Particle Duality / Energy of a Photon

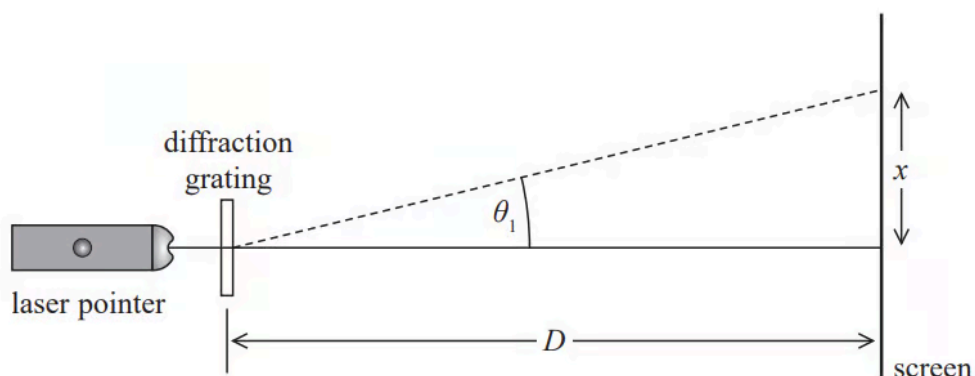
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Total Marks

/32

- 1 (a) Light from a laser pointer was passed through a diffraction grating. The light was perpendicular to the diffraction grating as shown. A diffraction pattern was produced on a screen.



The distance between the first order maximum and the central maximum of the diffraction pattern was x . The distance between the diffraction grating and the screen was D .

Distance x was measured to be 0.500 m with a metre rule. The wavelength of light λ_1 from the laser pointer was 650 nm.

The laser pointer was replaced with one that produced light of a different wavelength. The new distance x was measured to be 0.400 m.

$$D = 1.45 \text{ m}$$

Calculate the wavelength λ_2 of the light emitted by the replacement laser pointer.

$$\lambda_2 = \dots\dots\dots$$

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(5 marks)

- (b) Explain one modification to this method that would decrease the uncertainty in the calculated value of λ_2 .

(2 marks)

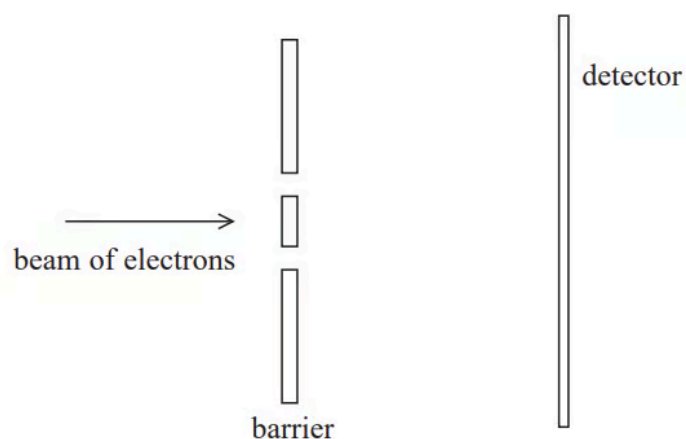
- (c) In another experiment, the light from the laser pointer was not quite perpendicular to the screen.

Explain how this would change the diffraction pattern produced on the screen.

(3 marks)

- 2 In 1965, Richard Feynman proposed a double slit experiment to investigate the wave properties of electrons.

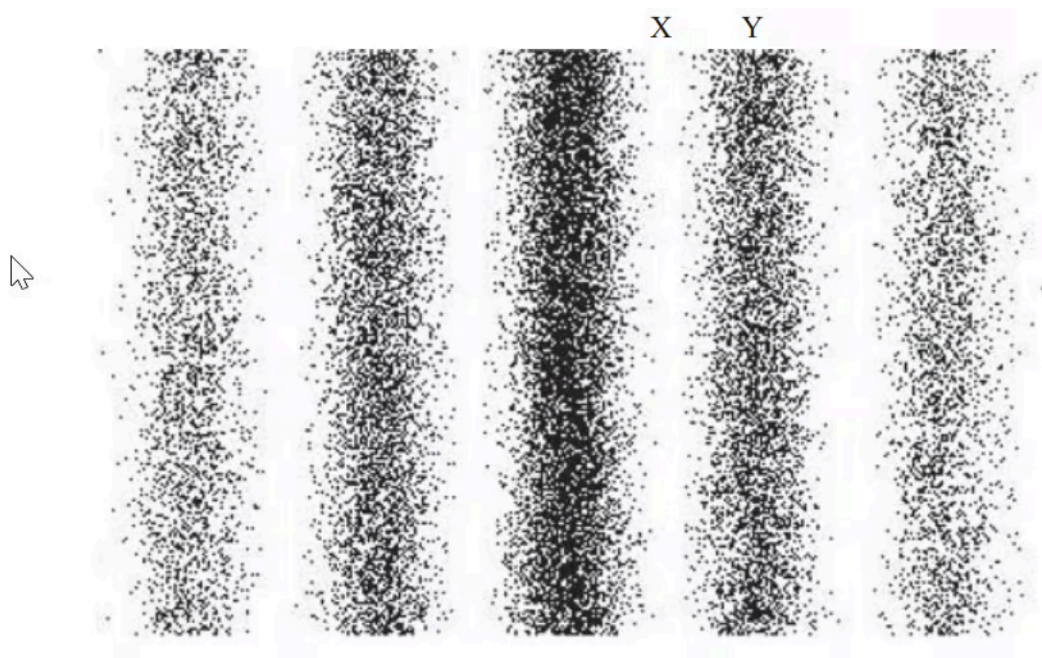
The experiment was later carried out using the arrangement shown.



A beam of electrons was directed at a barrier with two slits.

The detector recorded the positions where electrons arrived after passing through the slits.

The following pattern was obtained. Black dots represent points where electrons were detected. A band where electrons were not detected has been labelled X and a band where electrons were detected has been labelled Y.



The path difference for electrons arriving at band X from the separate slits was 2.5×10^{-11} m.

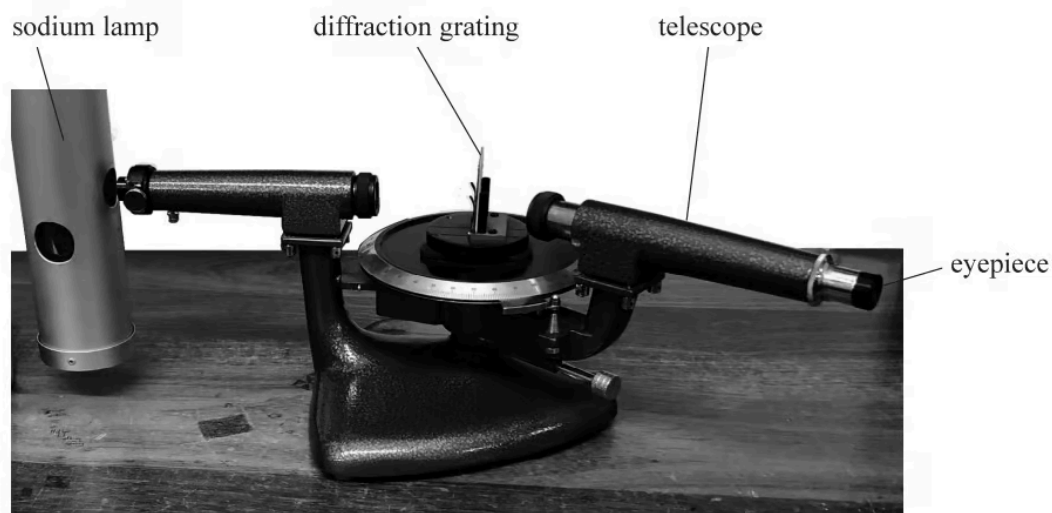
For electrons arriving at band Y the path difference was 5.0×10^{-11} m.

Explain why this pattern is observed when the electron energy is 9.6×10^{-17} J.

The electrons are travelling at non-relativistic speeds.

(6 marks)

3 (a) The photograph shows a school spectrometer.



The spectrometer allows parallel rays of light to be passed through a diffraction grating and the resulting angles of diffraction to be measured.

In the telescope, light from the grating is focused to make a real image 16.7 mm in front of the eyepiece lens. The eyepiece lens then uses this real image as an object to produce a magnified virtual image for the observer.

Calculate the magnification produced by the eyepiece lens.

focal length of eyepiece lens = 17.9 mm

Magnification =

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(3 marks)

(b) The spectrometer and diffraction grating are used to analyse the light from a sodium lamp. In the sodium lamp, sodium is heated until it becomes a vapour and an electric current is passed through it. The vapour then emits light.

After the light passes through the diffraction grating a line spectrum is observed.

i) Explain why only certain wavelengths are observed.

(6)

ii) Diffraction gratings with the following spacings are available:

$d / 10^{-6} \text{ m}$	1.0	1.7	2.0	3.3
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Explain which would be the best spacing to use to measure the diffraction angle for the third order maximum for yellow light of wavelength 589 nm.

(3)

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(9 marks)

(c) The diagram shows some of the energy levels in a sodium atom.

— 0.00 eV

— -1.02 eV

== -1.39 eV

— -1.52 eV

— -1.95 eV

— -3.04 eV

— -5.14 eV

Add an arrow to the diagram to show the transition involved in the emission of yellow light of wavelength 589 nm.

Show your working below.

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(4 marks)